📘 SQL Queries in DBMS – Notes with Syntax and Examples

🔷 1. Data Definition Language (DDL)

Used to define and modify database structures (tables, schemas).

a) `CREATE` – Create new table

Syntax:

```
CREATE TABLE table_name (
  column1 datatype CONSTRAINTS,
  column2 datatype,
  ...
);
```

Example:

```
CREATE TABLE Students (
  ID INT PRIMARY KEY,
  Name VARCHAR(100),
  Age INT,
  Dept VARCHAR(50)
);
```

---

b) `ALTER` – Modify existing table

Syntax:

```
ALTER TABLE table_name
ADD column_name datatype;
```

Example:

```
ALTER TABLE Students
ADD Email VARCHAR(100);
```

---

c) `DROP` – Delete a table completely

Syntax:

```
DROP TABLE table_name;
```

Example:

```
DROP TABLE Students;
```

---

d) `TRUNCATE` – Remove all records, keep structure

Syntax:

```
TRUNCATE TABLE table_name;
```

Example:

```
TRUNCATE TABLE Students;
```

🔹 2. Data Manipulation Language (DML)

Used for inserting, updating, and deleting data.

a) `INSERT INTO` – Add new data

Syntax:

```
INSERT INTO table_name (column1, column2, ...)
VALUES (value1, value2, ...);
```

Example:

```
INSERT INTO Students (ID, Name, Age, Dept)
VALUES (1, 'Hasib', 22, 'CSE');
```

---

b) `UPDATE` – Modify existing data

Syntax:

```
UPDATE table_name
SET column1 = value1, column2 = value2
WHERE condition;
```

Example:

```
UPDATE Students
SET Age = 23
WHERE ID = 1;
```

---

c) `DELETE` – Remove records

Syntax:

```
DELETE FROM table_name
WHERE condition;
```

Example:

```
DELETE FROM Students
WHERE Age < 18;
```

🔹 3. Data Query Language (DQL)

Used to fetch data from the database.

a) `SELECT` – Retrieve data

Syntax:

```
SELECT column1, column2
FROM table_name
WHERE condition;
```

Example:

```
SELECT Name, Age
FROM Students
WHERE Dept = 'CSE';
```

SELECT with ALL columns:

```
SELECT * FROM Students;
```

---

4. Data Control Language (DCL)

a) `GRANT` – Give privileges

Syntax:

```
GRANT privilege
ON object
TO user;
```

Example:

```
GRANT SELECT, INSERT
ON Students
TO user1;
```

b) `REVOKE` – Take back privileges

Syntax:

```
REVOKE privilege
ON object
FROM user;
```

Example:

```
REVOKE INSERT
ON Students
FROM user1;
```

🔹 5. Transaction Control Language (TCL)

Manages changes made by DML.

a) `COMMIT` – Save changes

```
COMMIT;
```

b) `ROLLBACK` – Undo changes

```
ROLLBACK;
```

c) `SAVEPOINT` – Set a point to rollback to

```
SAVEPOINT sp1;
```

Example:

```
BEGIN;
UPDATE Students SET Age = 25 WHERE ID = 1;
SAVEPOINT sp1;
DELETE FROM Students WHERE Age < 20;
ROLLBACK TO sp1;
COMMIT;
```

🔹 6. Clauses in SQL

a) `WHERE` – Filters records

```
SELECT * FROM Students WHERE Age > 20;
```

b) `ORDER BY` – Sort results

```
SELECT * FROM Students ORDER BY Name ASC;
```

c) `GROUP BY` – Group similar data

```
SELECT Dept, COUNT(*) FROM Students GROUP BY Dept;
```

d) `HAVING` – Condition for grouped data

```
SELECT Dept, COUNT(*)
FROM Students
GROUP BY Dept
HAVING COUNT(*) > 2;
```

🔹 7. Joins (basic overview)

- **INNER JOIN:** Match in both tables
- **LEFT JOIN:** All from left + match from right
- **RIGHT JOIN:** All from right + match from left
- **FULL JOIN:** All records from both, with NULLs for no match

Example:

```
SELECT s.Name, d.DeptName
FROM Students s
INNER JOIN Department d
ON s.Dept = d.DeptID;
```

☑ Summary Table

Type	**Command**	**Purpose**
DDL	`CREATE`, `ALTER`, `DROP`, `TRUNCATE`	Structure
DML	`INSERT`, `UPDATE`, `DELETE`	Data manipulation
DQL	`SELECT`	Data querying
DCL	`GRANT`, `REVOKE`	Permissions
TCL	`COMMIT`, `ROLLBACK`, `SAVEPOINT`	Transactions

Let me know if you want:

- **Joins in detail with diagrams**
- **ER model to relational schema conversion**
- **Normalization (1NF, 2NF, 3NF) explanation with examples**